

From Analogue to Calibration: A Reproducible SAT Geometry Workflow

Nathan McKnight

SAT/H(s)H research archive

Caliper

STORYO Precision Office — Keeper of the Analogue Ledger

(Dated: September 28, 2026)

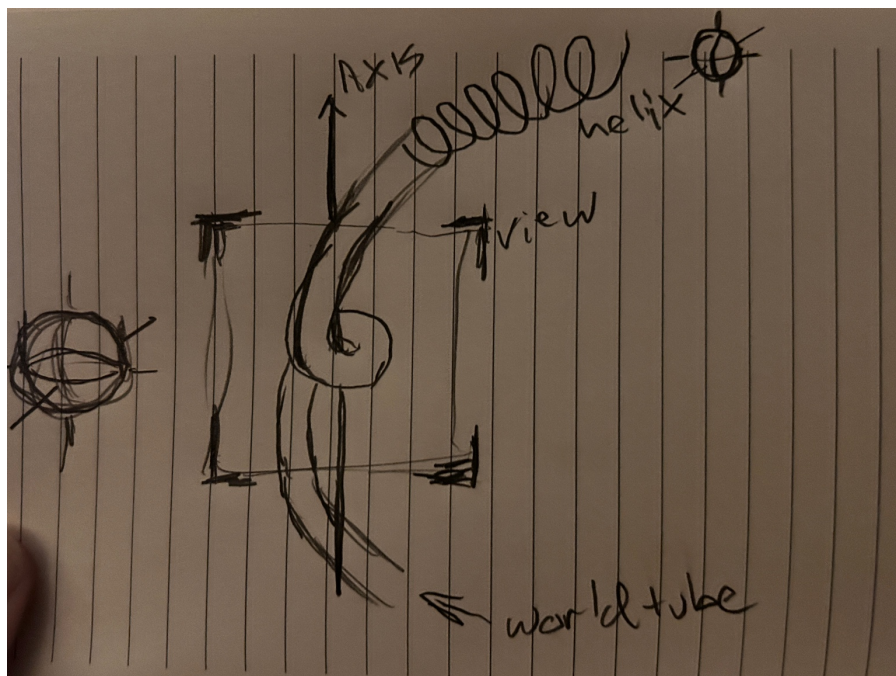


FIG. 1. Source analogue. The working rule was to extract constraints from the sketch rather than replace it with an illustrative interpretation.

Abstract

This note documents a working method for turning a rough SAT/H(s)H geometric sketch into a reproducible, inspectable, and eventually calibratable computational object. The method deliberately separates source analogue, model geometry, path-length accounting, physical normalization, and display. The central rule is simple: compute the undistorted object first; distort only the observation map, and record the map. The present build uses an arc-length-parameterized backbone, a parallel-transport frame, one or more rotating transverse tracers, recursive wrapping, explicit path-length ratios, and a same-path-speed constraint. The procedure is presented as a how-to and derivation ledger, not as evidence that the current geometry is a validated physical model.

I. WHAT WORKED

The successful part of the build was not a particular rendering style. It was the change in method.

We began with a hand-drawn analogue containing only a few geometric commitments: a backbone/worldtube, a localized curl, a helical continuation, an axis/view relation, and a transverse end view. Figure 1 is useful precisely because it is under-specified. It does not pretend to know more than the geometric idea.

An early cleaned line drawing, Fig. 2, was visually clearer but taught an important lesson: cleanliness is not the same as fidelity. A polished image can silently add geometry. The useful transition was therefore from image interpretation to explicit construction.



FIG. 2. Cleaned reconstruction. It clarified some spatial relations, but also demonstrated why the source image cannot be allowed to determine unrecorded geometry.

NO METAPHORS: ONLY ANALOGUES. An analogue earns its place by preserving a relation that matters: topology, ordering, symmetry, scale, constraint, transition, or dependency.

II. METHOD IN ONE LINE

analogue $\rightarrow$ backbone $\rightarrow$ frame $\rightarrow$ wrap $\rightarrow$ measure $\rightarrow$ calibrate $\rightarrow$ display

(1)

The order matters. In particular, display is last.

III. STEP 1: PRESERVE THE SOURCE AND ISOLATE PRIMITIVES

Do not begin by asking what the object “looks like.” Ask which distinctions the source actually commits to. For the present build those primitives were:

- a parent centerline $\mathbf{C}(s)$;
- a local tangent direction $\mathbf{T}(s)$;
- a local plane normal to $\mathbf{T}$;
- a transverse radius $R(s)$;
- an angular phase $\theta(s)$;
- one or more nested tracer paths;
- an observation direction that is not part of the object itself.

Everything else — glow, opacity, apparent line width, black core, perspective compression, and magnification — belongs to the observation map unless separately promoted to the model.

IV. STEP 2: PARAMETERIZE BY ARC LENGTH

Given sampled points $\mathbf{P}_i$, define cumulative path length

$$s_0 = 0, \quad s_i = s_{i-1} + \|\mathbf{P}_i - \mathbf{P}_{i-1}\|. \quad (2)$$

Then resample all three coordinates on a uniform s grid. This avoids confusing arbitrary drawing/sample density with physical or geometric density.

Listing 1. Arc-length resampling.

```
def cumulative_arclength(P):
    d = np.linalg.norm(np.diff(P, axis=0), axis=1)
    return np.concatenate([[0.0], np.cumsum(d)])

def resample_by_arclength(P, n):
    s = cumulative_arclength(P)
    t = np.linspace(0.0, s[-1], int(n))
    return np.column_stack([
        np.interp(t, s, P[:, i]) for i in range(3)
    ])
```

V. STEP 3: TRANSPORT A LOCAL FRAME

A curved backbone requires a transverse plane at every s . A naive Frenet frame can flip or become unstable near low curvature. We therefore transport an initial normal as minimally as possible from tangent to tangent.

Let $\mathbf{N}_1(s)$ and $\mathbf{N}_2(s)$ span the plane orthogonal to $\mathbf{T}(s)$. Between adjacent samples, rotate the old normal by the minimal Rodrigues rotation that takes $\mathbf{T}_i$ to $\mathbf{T}_{i+1}$, then re-orthogonalize.

This makes the rotating arm a genuine three-dimensional local object rather than a screen-space stamp.

VI. STEP 4: TRACE THE ROTATING ARM

The central construction is

$$\mathbf{X}(s) = \mathbf{C}(s) + R(s) [\mathbf{N}_1(s) \cos \theta(s) + \mathbf{N}_2(s) \sin \theta(s)]. \quad (3)$$

For N turns distributed over backbone length L ,

$$\theta(s) = \theta_0 + 2\pi N \frac{s}{L}. \quad (4)$$

If a loop packet is localized, do not fake it by bending the render. Localize the phase law itself: plateau before, smooth rotation interval, plateau after.

Listing 2. Core wrap operator.

```
def wrap_curve(spine, turns, radius, phase0=0.0):
    _, N1, N2 = parallel_transport_frame(spine)
    s = cumulative_arclength(spine)
    u = s / s[-1]
    phi = phase0 + 2*np.pi*turns*u
    return spine + radius*(
        N1*np.cos(phi)[: ,None] +
        N2*np.sin(phi)[: ,None]
    )
```

VII. STEP 5: RECURSE WITH THE SAME OPERATOR

The first wrapped curve can itself become the next spine. No new geometric trick is required. The same operation is simply reapplied:

$$\mathbf{C}_0 \rightarrow \mathbf{C}_1 \rightarrow \mathbf{C}_2 \rightarrow \dots \quad (5)$$

Here denotes the current recursive helix-to-superhelix operation. The essential discipline is that each order is built from the previous order's actual coordinates and local frame, not drawn around its projection.

Figure 3 shows the resulting high-density regime. What looks like a luminous tube is a densely sampled continuous path around a darker parent trajectory.

VIII. STEP 6: KEEP MULTIPLE PATHS GENUINELY INDEPENDENT

A second path must receive its own backbone, transported frame, phase, radius, and path-length ledger. Merely offsetting or duplicating screen geometry is not sufficient. Figure 4 shows the two-path test used in this build.

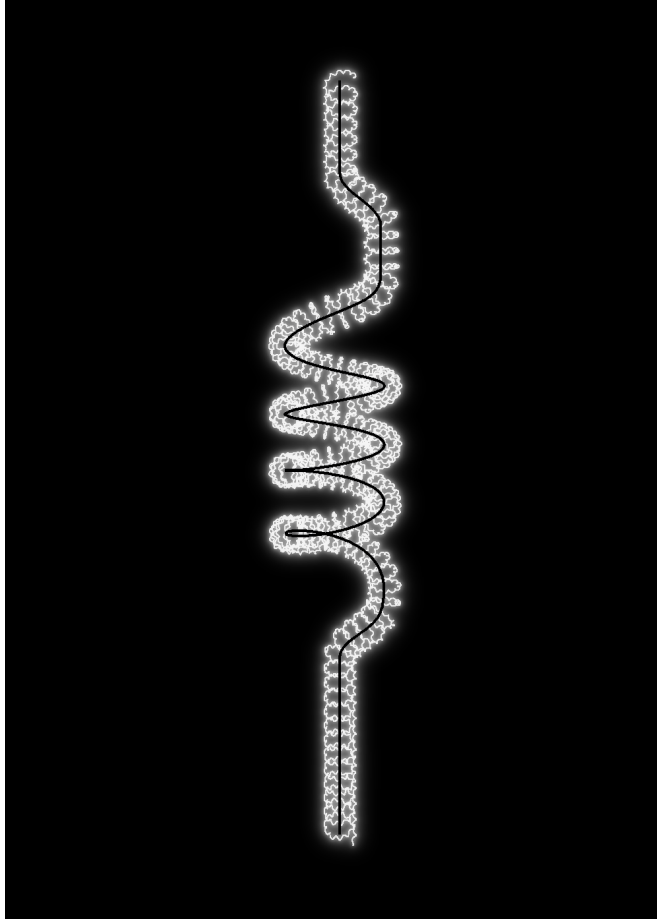


FIG. 3. Dense nested path. The apparent tube is an unresolved winding envelope at this display scale; it is not a replacement primitive.

IX. STEP 7: MEASURE PATH LENGTH BEFORE ASSIGNING SCALE

For any sampled curve $\mathbf{X}_i$, compute

$$L[\mathbf{X}] \approx \sum_i \|\mathbf{X}_{i+1} - \mathbf{X}_i\|. \quad (6)$$

The current dual-path build gave a macro-path/backbone ratio of approximately 7.72 and a fully nested path/backbone ratio of approximately 10.11. That immediately showed that the visual object was still far from the hypothesized dense-path regime.

The correction was not to “make it look denser.” We solved the inverse problem: with backbone and macro radius fixed, how many turns produce a requested path-length ratio?

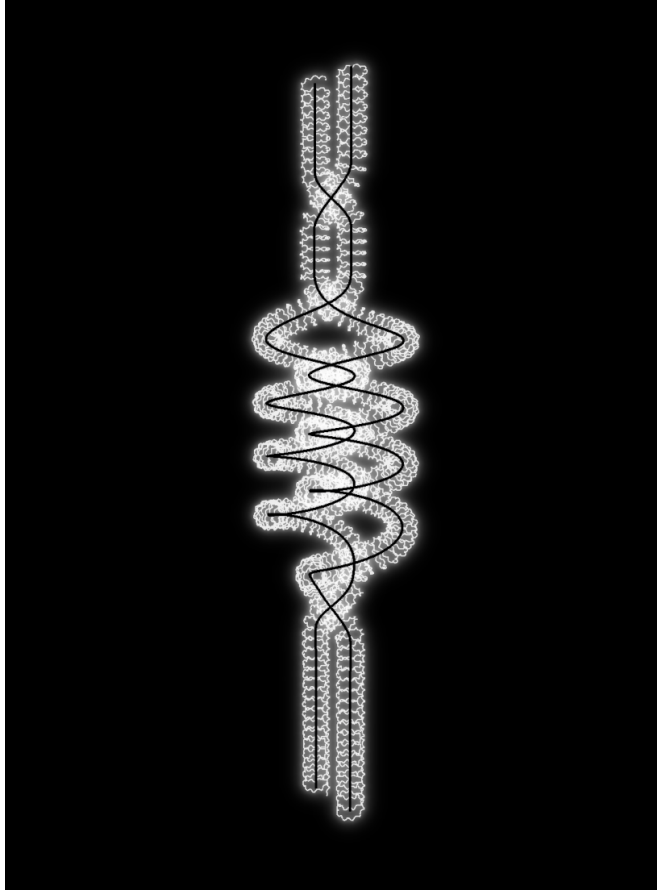


FIG. 4. Two independently constructed dense paths. Their geometry can later be coupled dynamically; in the present stage the important point is that they are not one rendered tube copied twice.

X. STEP 8: SOLVE WINDING DENSITY, DO NOT CHOOSE IT

For a straight ideal helix of radius R and axial pitch p per turn,

$$\eta = \frac{\ell_{\text{helix}}}{\ell_{\text{axis}}} = \sqrt{1 + \left(\frac{2\pi R}{p}\right)^2}. \quad (7)$$

For the curved numerical object we use direct arc-length integration and bisection instead of relying on the straight-helix approximation.

Figure 5 is the crucial visual lesson: above moderate winding density the cases become hard to distinguish by eye at this scale. That is a resolution fact, not evidence that the ratios are interchangeable.

TABLE I. Current inverse solutions with macro radius fixed at 0.20 model units.

Target ratio	Required turns	Same-path-speed parent advance
20:1	391.53	5.0%
50:1	982.00	2.0%
100:1	1979.00	1.0%
200:1	4086.58	0.5%

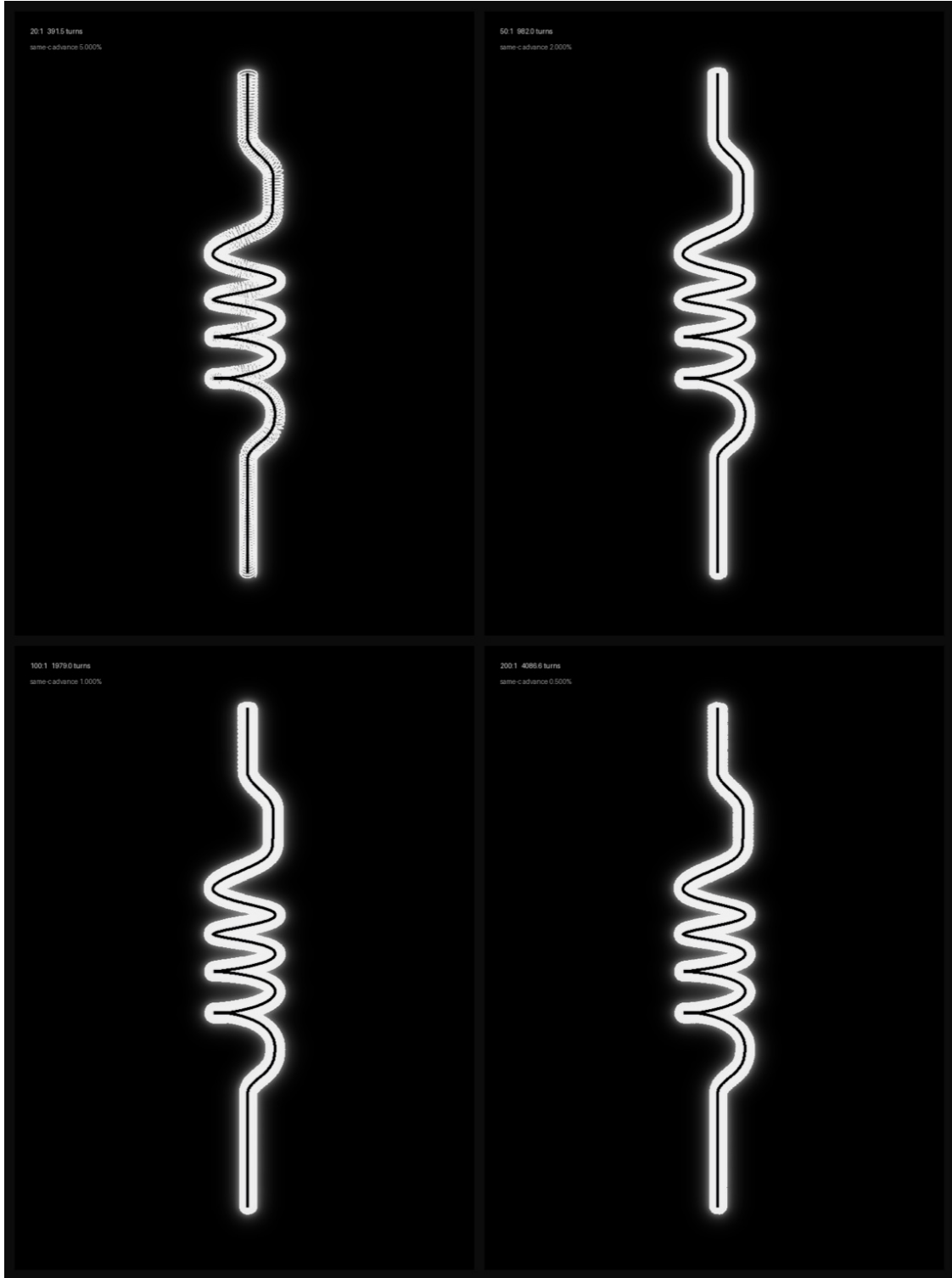


FIG. 5. 20:1, 50:1, 100:1, and 200:1 path-ratio sweep. High-density structures converge visually toward a tube at ordinary display resolution. The computed ratios remain distinct.

XI. STEP 9: ENFORCE EQUAL PATH SPEED EXPLICITLY

If two tracers each move at path speed c , a tracer traveling the longer wrapped path cannot occupy the same parent parameter s at the same elapsed time. The simultaneous parent-coordinate advance fraction is

$$f = \frac{L_{\text{parent}}}{L_{\text{child}}}. \quad (8)$$

Thus a 20:1 path ratio implies $f = 0.05$; a 200:1 ratio implies $f = 0.005$. The resulting “unsheathing” is therefore a consequence of path geometry plus the equal-speed rule, not an artistic offset.

This is also why a static picture is insufficient for the precision pass: the calculation must retain both spatial coordinates and the chosen temporal/proper-time parameterization.

XII. STEP 10: SEPARATE THE OBJECT FROM THE OBSERVATION MAP

The model should produce coordinates first:

$$\mathbf{X} = \mathbf{X}(s, t; \text{model parameters}). \quad (9)$$

Only then should a display transformation $\mathcal{V}$ be applied:

$$\text{pixels} = \mathcal{V}[\mathbf{X}]. \quad (10)$$

A figure should therefore carry a compact display ledger:

MODEL / PHYSICAL

length scale	= ...
time scale	= ...
transverse radius	= ...
phase rate	= ...
path speed	= c

DISPLAY

axial expansion	= 1.000x
-----------------	----------

radial expansion = 1.000x
time expansion = 1.000x
phase decimation = none

If the physical/model object is too dense to inspect, use one of the following without changing the stored geometry:

- higher sampling resolution;
- an optical magnifier registered to the same model coordinates;
- an explicitly declared axial or time expansion;
- an unwrapped (s, θ) companion view.

XIII. THE MAGNIFIER RULE

The magnifier should behave like a real lens over a higher-resolution master render. If the master is rendered at scale factor k and the lens center is (x_d, y_d) in display coordinates, the exact source center is

$$(x_m, y_m) = k(x_d, y_d). \quad (11)$$

The lens reveals a crop around that coordinate. It does not change phase, radius, pitch, or time. Optical zoom and structural expansion must therefore be separate controls.

XIV. WHAT THE FAILED PASSES TAUGHT US

The rejected stages are part of the method, not clutter to be erased.

1. **Illustration drift.** Cleaning the wrong generated geometry made it prettier but less faithful.
2. **Ribbon drift.** Turning the tracer into a ribbon added an unsupported primitive.
3. **Tooltip drift.** Screen-space stamps made three-dimensional local structure look flatter and changed apparent scale.
4. **Pearl drift.** Discrete luminous markers made a continuous dense path read as beads instead of a tube-like envelope.

5. **Aesthetic density.** Choosing winding by appearance concealed the path-length ratio. Solving the ratio removed that freedom.
6. **Resolution equivalence.** Once 50:1–200:1 look similar, the render alone cannot tell them apart. The ledger must.

Beauty is a compression of Truth. A clean figure is useful only if the compression preserves the distinctions the calculation needs.

XV. PRECISION PASS AGAINST SAT AND MEASURED SCALE

The next pass should be deliberately blind with respect to the desired visual answer.

1. Select one SAT-defined identity in a precise $ct = w$ -style coordinate convention.
2. Translate each primitive in the present geometry into that formalism: parent path, transverse radius, phase, recursion order, and time parameter.
3. Apply the Hagalaz operation $\square$ only through its declared mathematical definition; no hand-adjustment of the render.
4. Select independent measured constants and benchmark quantities (CODATA or experiment-specific values as appropriate).
5. Perform dimensional normalization before inspecting the rendered result.
6. Calculate all coordinates and path lengths.
7. Compare the result against the other SAT/4DHH Lagrangians and selected outside formalisms as separate checks, not as tuning targets.
8. Write every transformation, assumption, unit conversion, and residual to a running ticker-tape ledger.
9. Send the same ledger and result directly to the website/review surface so the public presentation cannot silently diverge from the calculation.

The useful outcome is not necessarily agreement. A failed normalization, incompatible path ratio, or inconsistent mapping is a result because it constrains the model without requiring a visual interpretation.

XVI. SHOP-FLOOR RULES

- **Unclench. Divide. Sort.**
- **Truth before algebra.**
- **Geometry before notation.**
- **Math only after the relation is actually there.**
- **Do not clog the theory. Keep it GeomeTidy.**
- **If a distinction can be represented with less added structure, use less.**

A concise version of the workflow is therefore:

$$\boxed{\text{observe} \rightarrow \text{separate} \rightarrow \text{braid} \rightarrow \text{check} \rightarrow \text{math} \rightarrow \text{shape}.} \quad (12)$$

Appendix A: Reference implementation

The separate file `geometry_engine.py` distributed with this note contains the complete minimal engine used by the current methodology: arc-length re-sampling, tangent calculation, parallel transport, localized phase, recursive wrapping, path metrics, same-path-speed accounting, and numerical inversion for target path ratios.

The essential inverse routine is:

```
def solve_turns_for_length_ratio(backbone, radius,
                                target_ratio,
                                lo=1.0, hi=10000.0,
                                iterations=40):
    Lb = cumulative_arclength(backbone)[-1]

    def ratio(turns):
        n = int(max(30000, turns*350))
        b = resample_by_arclength(backbone, n)
        h = wrap_curve(b, turns, radius)
        return cumulative_arclength(h)[-1] / Lb
```

```
for _ in range(iterations):
    mid = 0.5*(lo+hi)
    if ratio(mid) < target_ratio:
        lo = mid
    else:
        hi = mid

turns = 0.5*(lo+hi)
return turns, ratio(turns)
```

Appendix B: Reproducibility checklist

Before treating any new render as a model result, verify:

1. source analogue preserved;
2. coordinate convention stated;
3. parent path sampled by arc length;
4. local frame construction stated;
5. phase law stated;
6. all recursion orders separately recoverable;
7. actual path lengths reported;
8. path-speed rule stated;
9. unit normalization reported;
10. display transforms reported;
11. magnification distinguished from structural expansion;
12. failed/blocked alternatives retained in the ledger.